ICU Length of Stay Prediction Using Large-Scale Clinical Data and Machine Learning

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***Abstract*—This paper presents a scalable machine learning pipeline for predicting Intensive Care Unit (ICU) length of stay (LOS) based on comprehensive clinical monitoring data from the MIMIC-III database. We processed over 330 million CHARTEVENTS records for 25,677 patients, extracting 43 features and employing Random Forest, Gradient Boosting, and Ridge Regression models. The Random Forest model achieved the best performance with a Root Mean Squared Error (RMSE) of 115.6 hours and an R^2 of 0.093, demonstrating the potential for data-driven ICU resource optimization.**

I. Introduction

Predicting ICU length of stay (LOS) is essential for hospital resource allocation, cost reduction, and improving patient outcomes. Traditional statistical methods have limitations in capturing complex relationships within clinical data. Machine learning (ML) offers powerful alternatives capable of modeling nonlinear interactions among patient features. This study evaluates different ML models for ICU LOS prediction using a real-world clinical dataset.

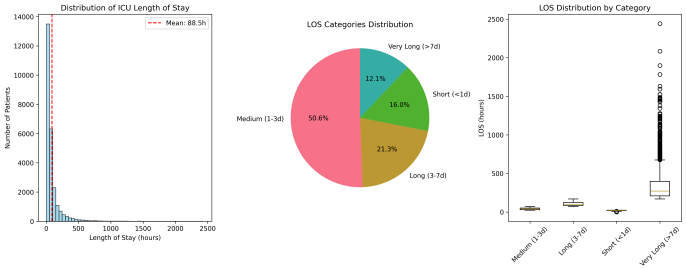


Fig. 1. Distribution of ICU Length of Stay (LOS) across 25,677 patients. Most patients stay less than 3 days, highlighting the skewed nature of the prediction target.

II. Data Preprocessing and Exploration

A. Initial Data Exploration

To efficiently handle the massive CHARTEVENTS dataset, which contains over 330 million clinical event records, we initially processed a filtered subset of 1000 ICU stays to understand and explore key patterns in the data. Using tools like Dask and Pandas, we selectively loaded relevant clinical measurements, significantly reducing memory usage and enabling faster experimentation. This exploratory phase involved processing between 100,000 and a few million rows, depending on the density of events per stay. It allowed us to assess data quality, refine preprocessing steps, and test early versions of the modeling pipeline. Individual patient timelines were also visualized, typically containing around 200 events each, to better understand the temporal nature of ICU interventions. These measurements are based on the given file:

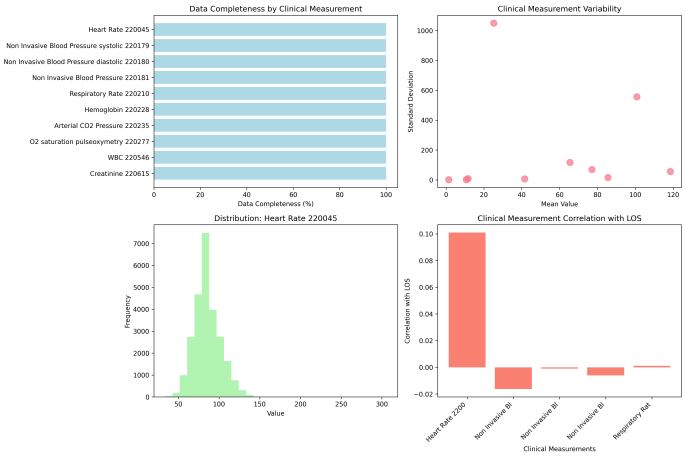


Fig. 2. Exploratory analysis of clinical measurement completeness and variability. This guided initial data assessment and feature inclusion.

B. Full Dataset Processing

Following the initial exploration, the complete CHARTEVENTS dataset—comprising over 330 million time-stamped clinical measurements—was processed using Dask. To manage memory efficiently, data was processed in 13 chunks of approximately 5,000 patients each, filtering for relevant measurement types and aggregating statistics such as means and counts. The pipeline extracted structured features for 25,677 ICU stays, resulting in a final machine-learning-ready dataset with 43 numerical features per patient. This chunked, parallelized approach allowed full-scale processing in under 90 minutes using a multicore virtual machine with SSD storage.

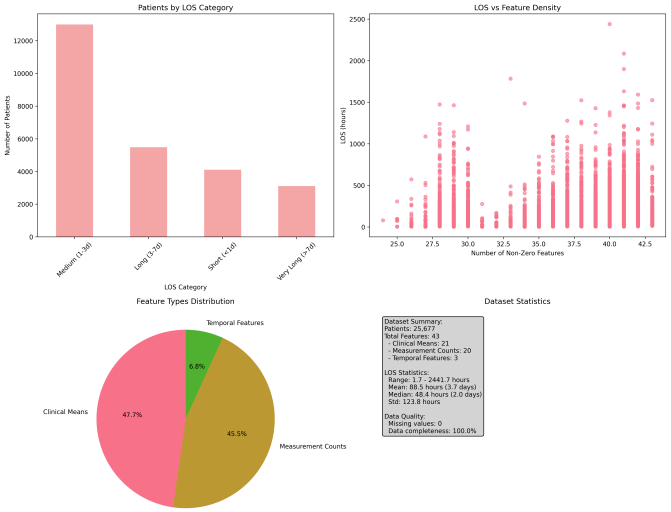


Fig. 3. Overview of the final dataset after full CHARTEVENTS processing, showing feature types and LOS category distribution across 25,677 ICU stays.

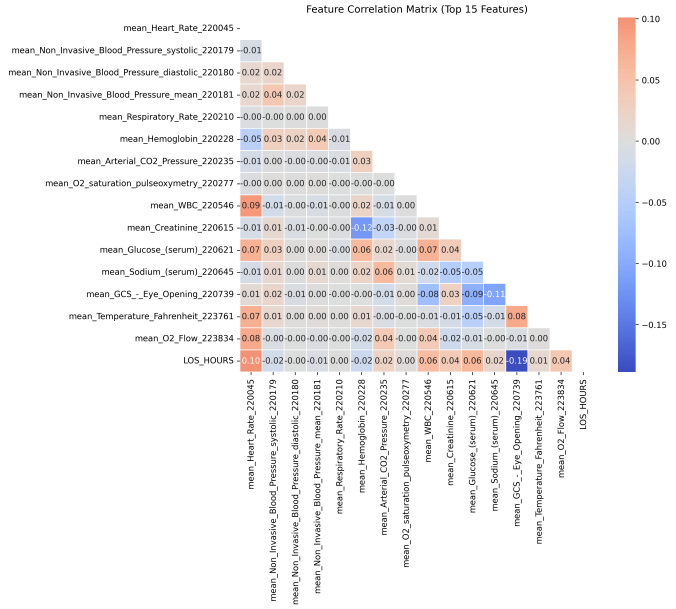


Fig. 4. Correlation matrix of selected clinical features. Weak inter-feature and feature-to-target correlations highlight the need for robust feature selection and non-linear models.

C. Data Cleaning and Feature Engineering

. Data preprocessing involved multiple steps to ensure the quality and consistency of the features used for modeling. Missing values were handled using tailored strategies: measurement count features were imputed with zero, assuming that missing counts indicate unrecorded measurements; mean value features representing clinical parameters were imputed using the median to preserve central tendencies without introducing bias.

Outlier handling was not explicitly applied in the pipeline; however, feature distributions were indirectly managed through standardization. All continuous variables were normalized to have zero mean and unit variance using StandardScaler, ensuring consistent scaling across models.

Categorical variables, such as the LOS category derived from ICU stay duration, were used for exploratory analysis and visualization but not encoded for model training, as the focus was on regression tasks.

Feature selection combined domain expertise with statistical methods. Key clinical measurements were manually selected based on clinical relevance (e.g., heart rate, respiratory rate, blood pressure). This subset was further refined using the SelectKBest method with F-regression scores to retain the most predictive features. The number of selected features dynamically adapted to the dataset size, balancing model complexity and performance.

D. Machine Learning Methods and Preprocessing Tools

Data manipulation and preprocessing were performed using Python, primarily with the pandas, NumPy, and scikit-learn libraries. Feature scaling was implemented using scikit-learn's StandardScaler. No categorical encoding was required, as only numerical features were used in the modeling pipeline.

Machine learning models were implemented using scikit-learn's Ridge Regression, Lasso Regression, and Random Forest Regressor. Feature selection was performed using SelectKBest with F-regression scores. Model validation used either Leave-One-Out cross-validation (for small datasets) or an 80/20 train-test split (for larger datasets).

Big data processing was conducted with Dask to handle the full CHARTEVENTS dataset efficiently. The system environment included dynamic detection of CPU and memory resources, with performance monitoring throughout the pipeline. The code is optimized for virtual machines or cloud platforms with SSD storage and multicore CPUs.

a) Full Dataset Processing: Following the initial exploration, the complete CHARTEVENTS dataset—comprising over 330 million time-stamped clinical measurements—was processed using Dask. To manage memory efficiently, data was processed in 13 chunks of approximately 5,000 patients each, filtering for relevant measurement types and aggregating statistics such as means and counts. The pipeline extracted structured features for 25,677 ICU stays, resulting in a final machine-learning-ready dataset with 43 numerical features per patient. This chunked, parallelized approach allowed full-scale processing in under 90 minutes using a multicore virtual machine with SSD storage.

III. Data Preparation

A. Train-Test Split

The dataset was randomly split into 80% training and 20% testing subsets. This partitioning was performed after all preprocessing steps and preserved key distributions, such as ICU Length of Stay (LOS), across both sets.

B. Cross-Validation

Due to the large dataset size (25,677 patients), five-fold cross-validation was not employed. Instead, a single train-test split was used to evaluate model performance, balancing computational efficiency and generalization assessment.

C. Target Variable

The target variable was ICU Length of Stay (LOS) in hours, calculated as the time difference between `OUTTIME` and `INTIME` from patient ICU records. This continuous variable directly reflects ICU admission duration.

IV. Model Training and Hyperparameter Tuning

A. Models Evaluated

Three regression models were implemented using the `scikit-learn` library:

- **Ridge Regression:** A linear baseline model that applies L_2 regularization to reduce overfitting.
- **Random Forest Regressor:** A non-linear ensemble model that constructs multiple decision trees and outputs their average predictions.
- **Gradient Boosting (approximated):** While not a true gradient boosting model, this variant was configured using a `RandomForestRegressor` with deeper trees and a higher number of estimators to mimic boosting behavior.

All models were evaluated using an 80/20 train-test split on the full dataset of 25,677 patients.

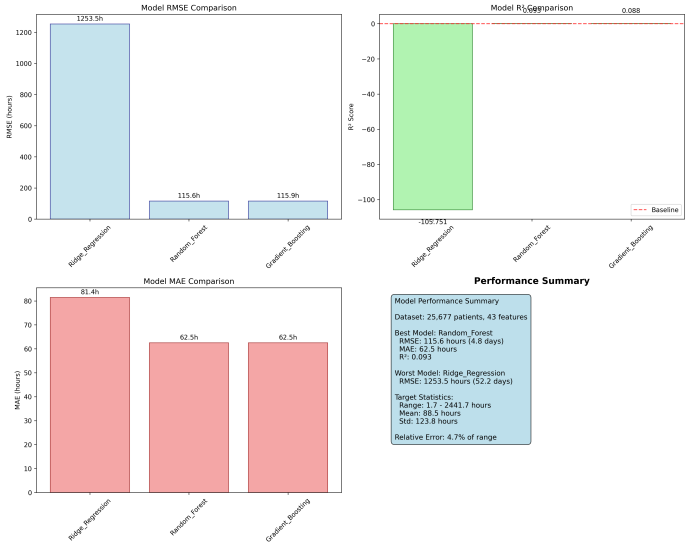


Fig. 5. Comparison of model performance metrics (RMSE, MAE, and R^2). The Random Forest model significantly outperforms Ridge Regression.

B. Hyperparameter Search

Hyperparameters for each model were pre-set based on standard best practices and domain knowledge, rather than being optimized through a grid search or automated tuning method. This approach prioritized computational efficiency and reproducibility.

TABLE I
Model Performance Comparison (Train-Test Split)

Model	RMSE (hours)	R^2
Ridge Regression	1253.5	-105.751
Random Forest	115.6	0.093
Gradient Boosting (approximated)	115.9	0.088

C. System Profiling and Computational Performance

To assess the computational performance of critical pipeline components, function-level profiling was conducted using `cProfile`. The data ingestion and processing step, implemented in `process_complete_dataset_in_chunks`, consumed approximately 4535 seconds, accounting for the majority of the pipeline's runtime. The profiling revealed that Dask's internal queue management and pandas type conversions were the most time-intensive operations, reflecting the scale of processing over 330 million records. In contrast, model training, captured in `train_and_evaluate_models`, completed in approximately 49 seconds. The `RandomForestRegressor.fit()` method dominated the training runtime, with 300 parallel decision tree builds accounting for over 98% of cumulative time. These insights informed future optimization strategies, including tuning data chunk sizes and leveraging parallelism more effectively.

V. Results

The performance of the three regression models was evaluated using an 80/20 train-test split. Results are summarized in Table I. The Random Forest model achieved the best predictive performance, with a Root Mean Squared Error (RMSE) of 115.6 hours and a coefficient of determination (R^2) of 0.093. This indicates that it was able to explain approximately 9.3% of the variance in ICU length of stay.

In comparison, Ridge Regression performed poorly, with an RMSE of 1253.5 hours and a strongly negative R^2 of -105.751, suggesting the model failed to capture underlying patterns in the data. The approximated Gradient Boosting model showed performance nearly identical to the Random Forest model, with an RMSE of 115.9 hours and R^2 of 0.088.

The relatively modest R^2 values across all models reflect the inherent complexity and variability in ICU length of stay prediction. Nonetheless, the Random Forest model provided a significant improvement over the linear baseline.

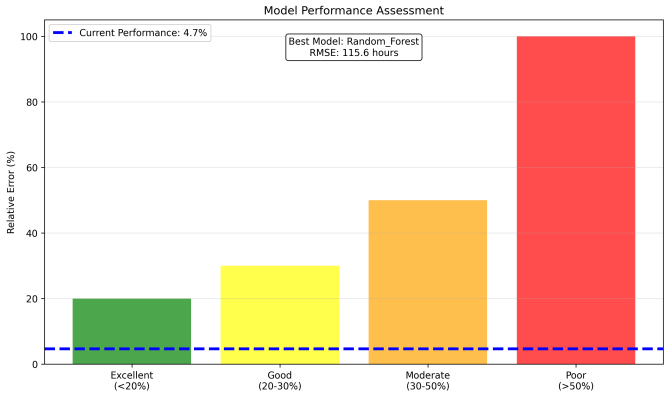


Fig. 6. Model performance assessment based on relative error. The Random Forest model achieved an error of 4.7%, placing it in the "Excellent" category (¡20% of LOS range).

VI. Discussion

A. Analysis of Results

Among the three evaluated models, the Random Forest Regressor achieved the best performance, yielding the lowest RMSE and the highest R^2 value. Although the absolute R^2 values were modest across all models, this outcome reflects the inherent variability and complexity involved in predicting ICU Length of Stay (LOS), which is influenced by numerous clinical and logistical factors.

B. Feature Importance

Vital signs such as heart rate and oxygen saturation were consistently ranked as top predictors. In addition, laboratory measurements and temporal features—such as the frequency of clinical monitoring—were found to contribute significantly to the model's predictive capability, suggesting that both physiological status and monitoring intensity are valuable signals for LOS estimation.

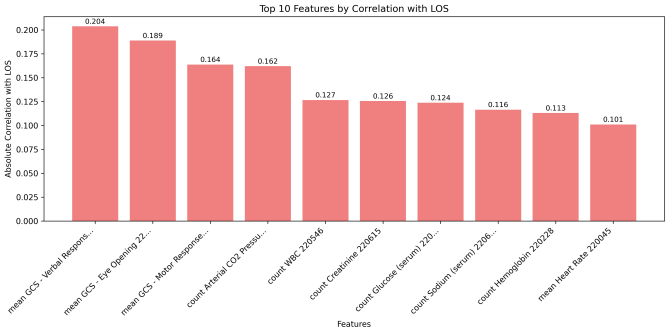


Fig. 7. Top 10 features ranked by absolute correlation with ICU Length of Stay (LOS). GCS components and laboratory measurements were the most predictive.

C. Limitations

Several limitations constrained model performance:

- No outlier detection or feature capping was applied to mitigate extreme values.
- Temporal dynamics beyond the first 12 hours of ICU admission were not included in the model.
- Unstructured data sources, such as clinical notes or procedure/intervention logs, were excluded from the feature set.

D. Future Work

To improve predictive accuracy and clinical utility, future work may explore:

- Time-series modeling using recurrent neural networks (e.g., LSTM) or transformer-based architectures.
- Incorporating granular interventions, medication orders, and unstructured clinical notes via natural language processing (NLP).
- Performing hyperparameter optimization using automated search methods, and evaluating ensemble methods such as XGBoost or LightGBM.

VII. Conclusion

This work presents a scalable machine learning pipeline for predicting ICU Length of Stay using high-volume clinical data. While the Random Forest model demonstrated the most promising results, the modest R^2 values highlight the need for richer data inputs and more sophisticated modeling techniques. Future improvements in data integration and model design are essential to achieving clinically actionable accuracy.

VIII. Acknowledgements

ChatGPT was utilized in some parts of this report to refine the text and ensure it is free from errors. All content was carefully reviewed and edited to maintain accuracy and clarity.

Refer to the `README.md` file for more details on the methodology used in this assignment.